

An **algorithm** is a **step-by-step procedure** used to solve a problem or perform a task. It is the **logical foundation** of programming and is usually written **before coding**.

Definition

👉 An algorithm is a **finite set of well-defined steps** that takes **input**, processes it, and produces **output**.

Characteristics of a Good Algorithm

1. **Input** – Takes zero or more inputs
 2. **Output** – Produces at least one output
 3. **Definiteness** – Each step is clear and unambiguous
 4. **Finiteness** – Terminates after a finite number of steps
 5. **Effectiveness** – Steps are simple and executable
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General Algorithm Structure

1. Start
 2. Input data
 3. Process data (logic/conditions/loops)
 4. Output result
 5. Stop
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Example 1: Algorithm to Add Two Numbers

1. Start
 2. Read two numbers A and B
 3. Compute $C = A + B$
 4. Display C
 5. Stop
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Example 2: Algorithm to Blink an LED (Embedded System)

1. Start
2. Initialize I/O port
3. Turn LED ON

4. Wait for delay
5. Turn LED OFF
6. Wait for delay
7. Repeat steps 3–6

Algorithm vs Program

Algorithm	Program
Step-by-step logic	Written in a programming language
Language independent	Language dependent
Easy to understand	Executable by machine

Algorithm Representation

- Plain English
 - Pseudocode
 - Flowchart
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Importance of Algorithms

- Helps in **problem analysis**
- Reduces programming errors
- Improves **efficiency**
- Makes programs easier to debug and modify

A **flowchart** is a **graphical representation of an algorithm**, using **symbols** to show the **sequence of steps** and **decision-making** in a process. It helps in **understanding, planning, and debugging** programs.

Common Flowchart Symbols

Symbol	Name	Purpose
 Oval	Terminator	Start / Stop
 Rectangle	Process	Calculation / Instruction

Symbol	Name	Purpose
◆ Diamond	Decision	Yes / No condition
▭ Parallelogram	Input / Output	Read / Display data
→ Arrow	Flow line	Direction of flow

General Flowchart Structure

1. **Start**
 2. **Input**
 3. **Process**
 4. **Decision** (if needed)
 5. **Output**
 6. **Stop**
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Example 1: Flowchart to Add Two Numbers (Text Form)

```

graph TD
    Start([Start]) --> Read[/Read A, B/]
    Read --> Process[C = A + B]
    Process --> Display[/Display C/]
    Display --> Stop([Stop])
  
```

Example 2: Flowchart for LED Blinking (Embedded System)

```

graph TD
    Start([Start]) --> Init[/Initialize Port/]
    Init --> LED[LED ON]
  
```

↓

Delay

↓

LED OFF

↓

Delay

↓

Repeat

Rules for Drawing Flowcharts

- Use **standard symbols**
- Flow should be **top to bottom or left to right**
- Avoid crossing flow lines
- Clearly label **decision paths**
- Use connectors for large flowcharts

Flowchart vs Algorithm

Algorithm	Flowchart
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Text-based	Diagram-based
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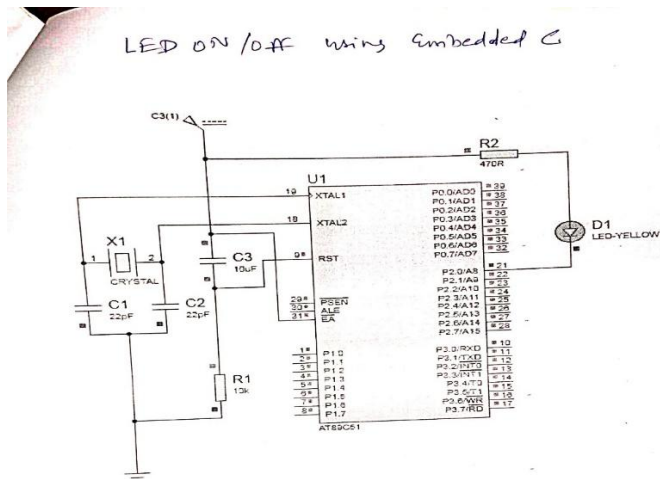
Easy to write	Easy to understand
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Less visual	More visual
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Importance of Flowcharts

- Simplifies complex logic
- Helps in error detection
- Improves communication
- Useful in exams and documentation
- Program to Blink an LED
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- This ALP program toggles an LED connected to **P1.0** at regular intervals using a delay loop.
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- ORG 0H
- MOV P1, #00H ; Set all pins of Port 1 to low (turn off LED initially)
- BLINK_LED:
 - MOV P1, #01H ; Set P1.0 to high (turn on LED)
 - ACALL DELAY ; Call delay subroutine
 - MOV P1, #00H ; Set P1.0 to low (turn off LED)
 - ACALL DELAY ; Call delay subroutine
 - SJMP BLINK_LED ; Repeat the loop
- DELAY:
 - MOV R0, #255 ; Set delay count
- DELAY_LOOP:
 - DJNZ R0, DELAY_LOOP ; Decrement and loop
 - RET ; Return from delay
- END
-



- **Program in C:-**

```
#include<reg51.h>
void main(void)
```

```
{
    unsigned int x;
    P1 = 0xFF;
    while (1)
    {
        x = P1;
        P0 = ~x;
    }
}
```

- **Result:**

Hence interfacing of thumbwheel switch to microcontroller 89C51 is studied successfully.

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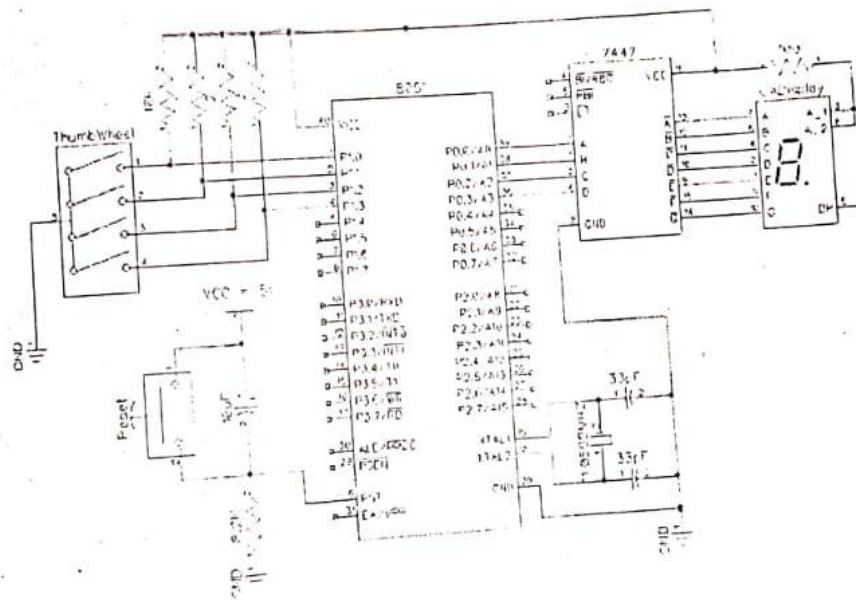
Sangola College, Sangola
Department Of Computer Science
Electronics Practical B.Sc. (E.C.S) – II

Ag. Expt. Interfacing of Thumbwheel Switch and 7 Segment display with 8051 Microcontroller

Aim: - To study interfacing of Thumbwheel Switch & 7 Segment Display to 89C51 Microcontroller.

Apparatus: - Microcontroller kit, thumbwheel Switch, 7 segment display, etc.

Circuit Diagram: -



Flow chart

